

N-DHT

Neuromorphic Distributed Hash Tables

A peer-to-peer routing fabric for an open internet — fast, learning-adaptive, and owned by no one.

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github.com/YZ-social/dht-sim

WHY WE ARE BUILDING THIS

"Those who would give up essential Liberty, to purchase a little temporary Safety, deserve neither Liberty nor Safety." — Benjamin Franklin

Privacy is the precondition for free expression and association. A communication channel that always passes through a trusted intermediary leaks the *fact* of the conversation even when the contents are encrypted. Knowing *who talked to whom, when, from where* is, on its own, a powerful surveillance signal — used commercially for advertising, used institutionally for law enforcement and intelligence.

End-to-end encryption is not enough. Encryption protects message contents but not metadata. As long as the routing fabric belongs to a single party — a server, a federation, an ISP — that party retains visibility into the network's *shape*, and that visibility is itself the asset.

A peer-to-peer routing fabric removes the custodian. No server, no privileged peer, no trusted coordinator: the routing layer becomes a participant-symmetric primitive in which no one party has a privileged view of the traffic.

***The DHT is the minimum viable substrate for that property.** Without it, every "decentralized" service is one trusted server away from being centralized again.*

WHAT'S BROKEN TODAY

The internet is supposed to be a peer-to-peer network. **In practice it isn't.** Routing, naming, content delivery, and increasingly identity all flow through a small number of intermediaries. Every alternative tried so far has fallen into one of three traps:

Approach	Examples	The trap
Trust a custodian	Twitter, Slack, Gmail, Cloudflare	Surveillance, censorship, single point of failure
Federate the custodianship	Mastodon, Matrix, ActivityPub	Server admins are the new custodians; problem moved, not solved
Decentralize but accept ~2-second routing	IPFS, BitTorrent, Ethereum devp2p	Real-time apps cannot live with multi-second lookups; ecosystem stalls

The result: twenty years of effort, no privacy-respecting alternative to the custodial internet has emerged at scale. Real applications — messaging, social, identity, real-time collaboration, federated AI — keep choosing custodians because the alternatives are too slow or too fragile.

The substrate that would unblock all of these — fast, peer-to-peer, custodian-free — has not existed. We built it.

WHAT WE BELIEVE

Six principles N-DHT is built on. Each is checkable in code; none is aspirational marketing:

- 1. No custodian.** No privileged node, no server, no trusted coordinator. The routing layer is participant-symmetric.
- 2. Fast enough to actually use.** Real-time means sub-second. ~2-second lookups disqualify a protocol from messaging, voice, social, gaming, and live collaboration. Speed is not optional.
- 3. Adaptive, not static.** The routing table learns from traffic. The protocol improves with use, not with manual tuning or operator intervention.
- 4. Built-in publish/subscribe.** Many-to-many delivery is a primitive, not a layer. Every modern application needs broadcast.
- 5. Browser-native.** Runs anywhere WebRTC runs. No special infrastructure, no servers to host, no operations to manage.
- 6. Public good.** Open source, no proprietary lock, no token, no exit strategy. The protocol is owned by no one because that is the only configuration in which it can mean what it claims to mean.

These are the values. The next slides are the engineering that delivers them.

THE PROTOCOL CLASS: N-DHT

N-DHT is a new class of distributed hash tables in which **routing tables learn from traffic**.

Every peer maintains a *synaptome* — a bounded set of weighted connections to other peers. Each connection carries a vitality score ($\text{weight} \times \text{recency}$). Successful routing reinforces the weights; unused connections decay. The routing table evolves toward the traffic the network actually carries, the way a cortical region's synaptic structure is shaped by the sensory input it processes.

Element	What it is
N-DHT	The protocol class — neuromorphic learning applied to routing
Synaptome	The vitality-scored routing table at each node (≤ 50 connections)
Vitality	$\text{weight} \times \text{recency}$ — Hebbian-reinforced score; the single admission gate
Axonal tree	Built-in publish/subscribe primitive; routed subscribe + reverse-path multicast
NH-1	Current implementation (Neuro-Homeostatic v1); 12 rules, 12 parameters

Lineage: twenty-five years of compounded routing research — Kademlia (2002) → Pastry / Tapestry (2001) → SCRIBE (2002) → eighteen named NX iterations → NH-1 (2026). Not invented from scratch; *consolidated* from existing work.

IT WORKS

1.18x

The Dabek 3δ analytical lower bound for any recursive $O(\log N)$ DHT — measured at 50,000 nodes.

Dabek et al. (NSDI 2004) proved that any recursive $O(\log N)$ DHT has a hard latency floor: total $\approx 3\delta$ where δ is the median pairwise one-way internet latency. The bound has stood unbeaten for twenty years.

Property	Result
Latency	NX-17 at 1.18x the floor (within 36 ms of the analytical optimum at 50K nodes); NH-1 at 1.28x; Kademlia at 2.65x and <i>worsens with scale</i>
Pub/sub delivery	100% baseline, 100% recovered after 5% churn — through routed re-subscribe alone, no replication, no gossip
Partition healing	Single-bridge connectivity dissolves into full network re-stitching via cooperative learning
Geography-independent	Strip the locality prefix entirely; learning still beats Kademlia by 26%

Performance is no longer the obstacle. The substrate exists.

WHAT THIS ENABLES

N-DHT is general-purpose infrastructure. Real-world applications across many domains:

Domain	What N-DHT enables
Privacy-respecting messaging	Signal-class infrastructure with no central server — no metadata trail, no custodian to compel
Decentralized identity	DID/ENS routing without identity intermediaries; verifiable credentials peer-to-peer
Social networks without platforms	Pub/sub fan-out at internet scale, no host, no algorithmic curator
Censorship-resistant publishing	Partition-tolerant content distribution; survives ISP filtering and regional outage
Real-time collaboration	Sub-second updates for documents, code, design — no WebSocket-server stack
Federated AI / agent coordination	Discovery + low-latency messaging for autonomous agents — without the hyperscaler API
IoT mesh and edge networks	Browser-native routing at the edge; no infrastructure to deploy
Decentralized commerce / settlement	Routing primitive for peer-to-peer markets, not custodial exchanges

The point is not any one of these. The point is the substrate that makes them all newly possible.

WHY THIS IS REAL, NOT SLIDEWARE

Five things separate N-DHT from the long tail of "decentralized" projects that never ship:

Twenty-five years of routing-research lineage	Kademlia → Pastry → Tapestry → SCRIBE → NX-1 through NX-17 → NH-1. Each generation addressed one measured failure mode in its predecessor. The current protocol is the consolidation of every working idea, with the failed ones explicitly archived.
Empirical validation, not promises	50,000-node open-source simulator (~25,000 lines of JavaScript). Every measurement in this deck is reproducible from a CSV in the public repository.
Peer-reviewed proof	12-page IEEE conference paper documenting the 3 δ -floor measurement. The first DHT in the published literature shown to operate at the analytical floor.
Three independent red-team analyses	Adversarial review of the protocol, the simulator, and the deployment story. Findings are public; deploy blockers are named, scoped, and time-estimated.
Everything open	Code, simulator, papers, red-team analyses, deck source, this pitch — all at github.com/YZ-social/dht-sim . No hidden component, no proprietary build.

This is engineering with measurements, lineage, and falsifiable claims. It is not a roadmap built from hope.

HOW IT GETS FUNDED

N-DHT is **open-source and non-profit**. The funding model is deliberate:

- **No tokens.** The substrate cannot be owned and cannot be speculated on. A protocol that requires you to buy in to participate is not a public good.
- **No proprietary fork.** The reference implementation is the canonical implementation. There is no commercial-only version waiting in the wings.
- **No exit strategy.** This isn't an investment; there is no liquidity event. The success state is *the protocol becomes infrastructure*, like TCP/IP, HTTP, DNS, and Linux did.
- **Funders shape direction, not ownership.** Funding pays for engineering runway. The substrate stays public. Funders' priorities can shape *what gets built next*, not *who owns it after*.

Why this funding model works: every layer of internet infrastructure that became universal — TCP/IP, DNS, HTTP, Linux, the open-source CA system, the IETF process itself — was funded into existence as a public good *before* private capital colonized the layers above. The pattern is well-tested. The alternative (token-funded, equity-funded, or hyperscaler-funded routing) has never produced a substrate with the properties this project requires.

The ask is for runway, not equity. Funders welcome at any size — foundations, mission-aligned corporations, individual technologists.

WHAT WE NEED

\$10M

24 months of engineering runway. Becomes infrastructure no one owns.

Allocation	%	What it buys
Production engineering	60%	2–3 senior protocol engineers; production hardening per the red-team Phase 1 deploy blockers; scale to 1 M-node simulator validation
Reference applications	25%	Three high-profile partner deployments — messaging, identity, agent coordination — demonstrating what becomes possible
Developer ecosystem	15%	SDKs, documentation, conference presence, integration partnerships with browsers and adjacent infrastructure

End state at 24 months: production-grade reference implementation, three reference applications on real networks, 100,000+ nodes in active deployment, SDKs that let third parties build on the substrate.



Runway, not equity. Public good, not platform. Open at every layer.

THE SUBSTRATE

Layer	Year	What it enabled
TCP/IP	1974	Internet routing
DNS	1983	Name resolution
HTTP	1990	The web
Linux	1991	An open operating system for the world

Each became universal because it was built as a public good — open, ownerless, available to anyone. Each was funded into existence by some combination of academia, government, foundation, and contributor labor *before* private capital arrived to build on top.

The fifth substrate is the peer-to-peer routing fabric the modern internet still does not have. It is what makes privacy-respecting messaging possible without a server. It is what makes decentralized identity possible without an intermediary. It is what makes federated AI possible without a hyperscaler. It is what makes social networks possible without a platform.

It needs to be funded the same way the first four were: as infrastructure that nobody owns and everybody can use.

That is what we are building. And what we are asking you to help build.